- Let f be a function from the set A to the set B. Let S be a subset of B. We define the **inverse image** of S to be the subset of A whose elements are precisely all preimages of all elements of S.
- We denote the inverse image of S by $f^{-1}(S)$, so $f^{-1}(S)=\{a\in A \mid f(a)\in S\}$.

44) Let f be the function from R to R defined by $f(x)=x^2$. Find

a) $f^{-1}(\{1\})$ Let, $f(x)=1$ $\therefore x^2=1 \Rightarrow x=+1,-1\in R$.

$\therefore f^{-1}(\{1\})=\{+1,-1\}$ □

b) $f^{-1}(\{x \mid 0<x<1\})$. Let, $f(x)=y$, where $y\in R$ and $0<y<1$.

$\therefore x^2=y$ $\therefore 0<x^2<1 \Rightarrow (x^2-1)<0 \Rightarrow (x+1)(x-1)<0$.

Two Cases: 1) $x+1>0$ and $x-1<0 \Rightarrow x>-1$ and $x<1$

2) $x+1<0$ and $x-1>0 \Rightarrow x<-1$ and $x>1 \Rightarrow$ Not possible.

$\therefore f^{-1}(\{x \mid 0<x<1\})=\{y \mid y>-1 \text{ and } y<1, y\neq 0\}$

c) $f^{-1}(\{x \mid x>4\})$. Let, $f(x)=y$, where $y\in R$ and $y>4$.

$\therefore x^2=y \Rightarrow x^2>4 \Rightarrow x^2-4>0 \Rightarrow (x+2)(x-2)>0$

$\Rightarrow$ Two Cases: 1) $x+2>0$ and $x-2>0 \Rightarrow x>-2$ and $x>2$ $\therefore x>2$

2) $x+2<0$ and $x-2<0 \Rightarrow x<-2$ and $x<2$ $\therefore x<-2$.

$\therefore f^{-1}(\{x \mid x>4\})=\{y \mid y>2 \text{ or } y<-2, y\in 4\}$

45) Let, $g(x)=\lfloor x \rfloor$. Find

(a) $g^{-1}(\{0\})$ Let, $g(x)=0$. $\therefore \lfloor x \rfloor=0$

By Property 1(a) from Table 1, $0\le x<0+1 \Rightarrow \boxed{0\le x<1}$

$g^{-1}(\{0\})=\{x \mid 0\le x<1\}$

(b) $g^{-1}(\{-1,0,1\})$ By Part (a), $g^{-1}(\{0\})=\{x \mid 0\le x<1\}$.

Let, $g(x)=1$. $\therefore \lfloor x \rfloor=1$

By Property 1(a) from Table 1, $1\le x<1+1 \Rightarrow \boxed{1\le x<2}$

Let, $g(x)=-1$. $\therefore \lfloor x \rfloor=-1$. By Property 1(a) from Table 1,

$-1\le x<-1+1 \Rightarrow \boxed{-1\le x<0}$